bm25_probabilistic_formulation (Initial Draft)

Domain: Information Retrieval | Document ID: bm25_probabilistic_formulation_v1
Specification Standard: IEEE Capstone Evaluation Corpus

Executive Overview:

Okapi BM25 scoring formula: inverse document frequency (IDF) saturation and sub-linear term frequency document length normalization.

Methodological Verification & System Lineage:

This specification is catalogued in IntentCloud for intent-aware hybrid retrieval.
Text chunks are indexed into 384-dimensional dense semantic vectors and 1,000,003-dimensional sparse feature-hashed token projections.
Candidate pools are unified using scale-invariant Reciprocal Rank Fusion (k=60).
Explainable citation attribution is extracted using cross-encoder cross-attention.